

B.Tech. BCSE497J - Project-I

**PneuNet: PNEUMONIA DETECTION USING
CONVOLUTION NEURAL NETWORK**

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Computer Science with Specialization in Data Science

by

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November 2025

DECLARATION

I hereby declare that the project entitled PneuNet: PNEUMONIA DETECTION USING CONVOLUTION NEURAL NETWORK submitted by me, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering* to VIT is a record of bonafide work carried out by me under the supervision of Prof. Sudhakar P.

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place : Vellore
Date : 12/11/25

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CERTIFICATE

This is to certify that the project entitled PneuNet: PNEUMONIA DETECTION USING CONVOLUTION NEURAL NETORK submitted by Swapnil Dwivedi (22BDS0269), Adarsh Tiwari (22BDS0350), **School of Computer Science and Engineering**, VIT, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering*, is a record of bonafide work carried out by him under my supervision during Fall Semester 2024-2025, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

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ACKNOWLEDGEMENTS

I am deeply grateful to the management of Vellore Institute of Technology (VIT) for providing me with the opportunity and resources to undertake this project. Their commitment to fostering a conducive learning environment has been instrumental in my academic journey. The support and infrastructure provided by VIT have enabled me to explore and develop my ideas to their fullest potential.

My sincere thanks to Dr. Jaisankar N, the Dean of the School of Computer Science and Engineering (SCOPE), for his unwavering support and encouragement. His leadership and vision have greatly inspired me to strive for excellence. The Dean's dedication to academic excellence and innovation has been a constant source of motivation for me. I appreciate his efforts in creating an environment that nurtures creativity and critical thinking.

I express my profound appreciation to Sendhil Kumar K S, the Head of the Data Science, for his/her insightful guidance and continuous support. His expertise and advice have been crucial in shaping the direction of my project. The Head of Department's commitment to fostering a collaborative and supportive atmosphere has greatly enhanced my learning experience. His constructive feedback and encouragement have been invaluable in overcoming challenges and achieving my project goals.

I am immensely thankful to my project guide, Prof. Sudhakar P, for his dedicated mentorship and invaluable feedback. His patience, knowledge, and encouragement have been pivotal in the successful completion of this project. My supervisor's willingness to share his/her expertise and provide thoughtful guidance has been instrumental in refining my ideas and methodologies. His/her support has not only contributed to the success of this project but has also enriched my overall academic experience.

Thank you all for your contributions and support.

Swapnil Dwivedi
Adarsh Tiwari

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List of Abbreviations

AI	Artificial Intelligence
ARDS	Acute Respiratory Distress Syndrome
CAD	Computer-Aided Design
CNN	Convolution Neural Network
CXR	Chest X-Ray
GB	Gigabyte
GPU	Graphics Processing Unit
GradCAM	Gradient-weighted Class Activation Mapping
JPEG	Joint Photographic Experts Group
MacOS	Macintosh Operating System
OpenCV	Open Source Computer Vision Library
PNG	Portable Network Graphics
QSCA	Quaternion Channel-Spatial Attention
RGB	Red, Green, and Blue
ROI	Region of Interest
SE	Squeeze and Excitation
SRS	Software Required Specification
TPU	Tensor Processing Unit
VGG	Visual Geometry Group
WHO	World Health Organization
YOLO	You Only Look Once

ABSTRACT

Pneumonia is an infection of the lungs that causes a major health problem globally and is still the main reason for death among children under five years old and in the elderly. People with the disease have an inflammation of their lungs accompanied by difficulty in breathing due to liquid accumulation. The problem may even result in a fatal outcome. Detecting pneumonia early is therefore crucial in order to administer treatment early enough and to have better chances of recovery. But a diagnosis using a chest X-ray is not done directly as it needs the expertise of a skilled radiologist. It is relatively slow, labor-intensive, and susceptible to errors of interpretation.

This work is about the construction of a system that can automatically diagnose pneumonia by using deep learning methods, mainly the CNN model. The CNN learns the difference between two kinds of images, i.e., chest X-ray images of Normal and Pneumonia patients, and then classifies the images into one of the two groups. In doing so, the CNN analyses visual features of the healthy lungs and the ones that are infected in the X-ray images. Along with these, the image pixels are also scaled down in value, the images are uniformly sized and cleaned of random noise in order to facilitate our model so that it can learn features well. Further, the authors of the paper resort to different techniques such as rotating, flip, and scaling to diversify the dataset and improve the power of our model for untested images from the test sets.

The CNN model that we built incorporates several strategies such as batch normalization, dropout, and early stopping to lessen overfitting and increase the model's precision. The model, after training and testing, is highly capable of identifying situations where the lungs are infected with pneumonia resulting from either bacteria or viruses. The performance of the system, gauged using metrics, like accuracy, and sensitivity, is solid and thus, the method can be considered as one of the aids in diagnostics.

The technique that is put forward in our work achieves automated image-based diagnosis and hence the turnaround time is reduced considerably; Thus, the solution is more affordable, and healthcare centers in underdeveloped areas without medical experts will benefit a lot from it. The task load of the radiologists is reduced, and the provision of some preliminary information on the screening facilitates clinical decision-making. To sum up, this research exhibits the potential of AI and computer vision to revolutionize medical imaging and thus be in line with other CAD systems, contributing to a faster diagnosis of respiratory diseases and enhancing healthcare quality.

Keywords

Pneumonia Detection, Convolutional Neural Networks, Deep Learning, Image Classification, Medical Imaging, Data Augmentation, Batch Normalization, Early Stopping, Computer-Aided Diagnosis, Chest X-ray, Healthcare AI.

CHAPTER 1

INTRODUCTION

1.1 Background

Pneumonia is a serious inflammatory condition of the lungs caused by bacteria, viruses, or fungi, that leads to the lungs' alveoli filling with fluid. Although chest X-rays are the main tool used to diagnose pneumonia, calling them correctly needs highly skilled radiologists and can be a very long process, which can also be affected by errors of human nature.

Driven by the advances in deep learning and artificial intelligence, the automated image analysis is considered a viable solution for the medical diagnostics. Convolutional Neural Networks (CNNs) have shown an impressive ability to identify complex patterns in medical images, thus, making disease detection both faster and more accurate.

This custom CNN model is created in the project and is merged with the pretrained model architectures to get the advantages of both transfer learning and domain-specific learning. While the pretrained models offer robust feature extraction capabilities, the custom CNN layers adjust the network to the specific features of pneumonia in a chest X-ray. Such a hybrid strategy allows the network to be more robust and accurate, thereby releasing an efficient and consistent system for detecting pneumonia.

1.2 Motivations

Pneumonia is still among the major factors leading to death all over the globe, regularly being the main cause of death for children aged under five and the elderly. As per findings released by the World Health Organization (WHO), there are newly confirmed cases in the millions every year, and a large number of deaths result from the late and/or inaccurate diagnosis. Though the chest X-ray is considered a primary examination method, visualizing these photos heavily relies on the expertise of the radiologist and the results may differ under various circumstances such as tiredness, amount of work, and lack of facilities.

This initiative's reasons are three major points:

- **Bridging the Healthcare Gap:** In places where medical resources are limited, the lack of skilled radiologists is typical. A computerized pneumonia identification module can be the medical staff's aid in decision making, thus the diagnosis becomes accessible and fast enough to patients even if they are far from the hospital.
- **Reducing Diagnostic Errors:** The X-ray manual reading can be influenced by the subjectivity of the person and errors may occur. In this way deep learning instruments especially CNNs can be used to make the system consistent and at the same time reduce the irregularities in the system, hence patient outcomes will be better.
- **Real-Time and Scalable Solutions:** Despite the accuracy, the conventional diagnostic approach takes a long time to perform and it's quite difficult to scale. The CNN-supported identification tool can deliver near to instant and precise results, thereby becoming feasible to implement in medical institutions, and m-health applications, as well as in the telemedicine sector for on-the-fly usage.

The project, through this concern, looks forward to producing a smart, cost-effective and

scalable solution that not only helps the work of the radiologists, but also makes a call to the healthcare infrastructure enabling it to save more lives because of early detection and the proper treatment of the diseases.

1.3 Scope of the Project

This project is limited to the binary categorization of chest radiographs, to be categorized as:

- Normal (lungs without disease)
- Pneumonia (lungs with infection)

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

R. Kundu et al. (2021) designed an ensemble-based deep learning framework integrating the architectures of GoogLeNet, ResNet-18, and DenseNet-121 for the identification of pneumonia from chest X-ray images. The ensemble model made use of a weighted average method to increase prediction reliability and overall system robustness. It was extremely successful in achieving a 98.8% accuracy rate on public datasets and also, the model exhibited very high sensitivity in detecting pneumonia cases. The paper pointed out that the ensemble learning approach solves the problem of overfitting in models and thereby, the generalization performance to the new data is improved.

In their paper, S. Singh et al. (2023) put forward the QCSA Network (Quaternion Channel-Spatial Attention) that combines quaternion residual blocks with spatial and channel attention. With this model, the authors were able to extract inter-channel dependencies from chest X-ray images which led to an improved classification of pneumonia. Their approach was able to reach an accuracy of 94.53%, which is higher than that of numerous CNN models. One of the points emphasized by the study was that quaternion representations have the ability to retain the rotational as well as the phase information which, in turn, makes the learning from medical images more exact and of higher efficiency.

In their review paper, R. Siddiqi and S. Javaid (2024) surveyed the deep learning methods for pneumonia recognition from chest X-ray files that were published between 2020 and 2023. Their paper explored CNN, DenseNet, MobileNet, and hybrid architectures and evaluated their performance, dataset diversity, and limitations. It offered a thorough comparative analysis of research results for the ChestX-ray14 and Kaggle pneumonia datasets, pointing out the necessity of data equilibrium and enhanced interpretability. The review paper finished with the statement that transfer learning and attention mechanisms are the main factors that lead to larger model performance.

In their research, H. Bhatt et al. (2023) proposed an ensemble CNN model that effectively combined several architectures in detecting pneumonia from chest X-ray images with increased accuracy. The paper extensively experimented with different CNN backbones like VGG16, ResNet50, and DenseNet121, and emphasized the advantage of merging their outputs. The ensemble technique far outperformed the single-model strategies leading to a considerable improvement of both recall and precision figures. Alongside, the paper also touched on the importance of the medical aspect of the issue in which it is crucial to decrease the number of false negatives in pneumonia diagnosis.

Y. Hedhoud et al. (2022) developed the CNN-XGBoost hybrid model with higher accuracy for the classification of pneumonia from chest X-rays. The hybrid system used CNN-based feature extraction followed by XGBoost application for decision making. The work showed that the system was very robust and interpretable with better performance metrics like F1-score and precision. The paper ended up saying that the hybrid fusion of deep and tree-based models provides a higher level of generalization for small medical datasets.

In an experiment with various transfer learning models like VGG16, ResNet50, InceptionV3, and DenseNet201, N. C. Kundur (2023) examined the effectiveness of these models in identifying pneumonia in chest X-ray images. To speed up the calculations and enhance the results, distributed training on TPU environments was used in the study. Out of all the tested models, DenseNet201 was able to distinguish the classes with an accuracy rate of over 96% and thus it was the best performer. The paper established that deep learning models with pre-trained weights in the CNN domain could achieve higher results than the conventional ways of image analysis and require lesser preprocessing of data.

T. Rahman et al. (2020) utilized transfer learning with AlexNet, ResNet18, DenseNet201, and SqueezeNet to identify pneumonia that is either bacterial or viral based on X-ray images. With 98% accuracy, the model made a significant leap in generalization capability as compared to conventional methods. The research pointed out how pre-trained CNNs can effectively extract knowledge from a vast number of images to help in medical diagnosis. Besides, it put forward a practical and easily reproducible method as a pipeline for healthcare facilities to onboard automated screening in a setting with fewer resources.

In their 2025 paper, R. Kailasam and S. Balasubramanian introduced a hybrid deep learning model that merges CNN classification and YOLO object detection to both identify and localize pneumonia in chest X-ray images. The model training resulted in an accuracy of 96.8% and the validation accuracy was 83%, so the model was good performance despite the variability of the dataset. Their method went beyond mere classification to also visually indicate the disease-infected areas of the lungs, thereby facilitating interpretability. Such a dual-stage architecture turned out to be very helpful for clinical decision support systems.

2.2 Research Gap

Limited Generalization of Models to Different Datasets

- The majority of CNN-based models (CheXNet, DenseNet, ResNet), have been developed by using data from single medical centers, such as Kaggle Pediatric CXR or ChestX-ray14.
- As a result, their accuracy is usually lowered when they are evaluated on data that come from different sources since the equipment used for imaging, the characteristics of patients, and the conditions of the acquisition may differ.
- **Gap:** There is a demand for models that can perform robustly across various hospitals and X-ray machines.

Insufficient Explainability

- Grad-CAM and Grad-CAM++ methods can show some visual interpretability of the working of the model, but in general, the majority of the models are "black boxes".
- Doctors require explicit explanations of model predictions to be able to rely on AI-assisted diagnosis.
- **Gap:** Creating significantly more transparent models that can not only highlight the areas but do so in a way that is clinically relevant and reliable.

Data Imbalance and Limited Dataset Size

- Several studies emphasize the problem of class imbalance (the number of pneumonia cases is less than the number of normal cases) as well as the limitation of annotated

datasets.

- Such a situation may result in biased predictions, overfitting, and low sensitivity of the minority classes.
- **Gap:** There is a requirement for strong augmentation techniques, generating synthetic data, or datasets from multiple institutions.

Real-Time Deployment Challenges

- The majority of current models have been trained and tested using high-end hardware. However, there is very little work done regarding their deployment on edge devices or in real-time clinical environments.
- **Gap:** Create lightweight, power-saving models that can perform real-time pneumonia detection in hospital settings or on point-of-care devices.

Multi-Disease Detection Capability

- The majority of the research works have a strong focus on binary classification, i.e., Differentiating between Normal and Pneumonia.
- From a clinical perspective, it is of utmost importance to be able to tell the difference between pneumonia and other lung diseases such as tuberculosis, COVID-19, and ARDS.
- **Gap:** Develop models that can recognize multiple classes which will provide broader diagnostic capabilities.

Limited Localization and Clinical Validation

- Some research works (for example, Kailasam et al., 2025) that utilized YOLO for ROI localization, while most classification models only infer image-level predictions.
- Clinical validation with radiologists and their involvement in the workflow is still very limited.
- **Gap:** Merging detection, localization, and explainability, along with thorough clinical trials.

2.3 Objectives

The main purpose of this work is to create and execute an automated deep learning system that can detect pneumonia within chest radiographs through model of a custom CNN and a ResNet152V2 pretrained network.

The specific objectives are:

- To create a custom CNN architecture that can understand spatial and texture-based features of the lung areas infected with pneumonia.
- To employ the ResNet152V2 pretrained model for transfer learning, benefiting from its deep feature extraction potential to augment model performance and lessen training complexity. ResNet152V2 is chosen over other pretrained architectures (such as VGG, Inception, MobileNet, or DenseNet) because it's very deep 152-layer design, identity-based skip connections, and pre-activation residual blocks allow superior gradient flow, improved convergence, and extraction of subtle radiographic patterns. The V2 variant provides better generalization and training stability than many alternative models, while being more memory-efficient than DenseNet and more accurate than lightweight

architectures like MobileNet for medical imaging tasks. These characteristics make ResNet152V2 particularly effective for detecting fine-grained features in chest X-rays, offering a stronger balance of accuracy, depth, and robustness compared to other models.

- To prepare and augment the dataset through scaling, image resizing, and dataset enhancement techniques to generalize better and prevent overfitting.
- To adjust the combined architecture to optimal hyperparameters that allow an equilibrium between performance efficiency and accuracy.
- To measure model performance through different evaluation indicators including accuracy, precision, F1-measure, recall, and confusion matrix.
- To design a powerful and trustworthy deep learning system able to facilitate the work of radiologists and support them in making fast and accurate diagnoses of pneumonia from X-ray images.

2.4 Problem Statement

Pneumonia ranks as a leading contributor to mortality worldwide, severely impacting mostly children and the elderly. The conventional way to diagnose the disease depends on the analysis of chest X-ray which is done by radiologists. However, this process takes a lot of time, is subject to different opinions, and may contain mistakes. Besides, a shortage of skilled radiologists is reported in many remote and less developed areas, thus the detection and treatment in time become a challenge.

A neural network method, especially a Convolutional Neural Network (CNN), has noticeably raised a medical image classification level, but still, the need for a local, trustworthy, and easy-to-use system, capable of recognizing pneumonia automatically in chest X-ray images, remains open. The main difficulties are the handling of various image qualities, achieving a high accuracy in classification, and providing an easily understandable output to be able to help the medical professionals in their clinical decision-making.

As such, an automated pneumonia detection system using image preprocessing, CNN-based classification, and real-time prediction is essential to be integrated with the medical care professionals to ease their work, lower the diagnosis time, and elevate the patients' well-being.

2.5 Project Plan

The project phases were designed to ensure a structured and efficient workflow that would take the team through the project from the initial planning stage all the way to the final documentation stage. Each phase was made up of specific tasks that had their timelines and deliverables specified.

2.5.1 Project Initiation & Planning

The project initiation and planning stage was primarily about pinpointing the project's scope, goals, and overall strategy so that the whole team would have the same understanding before the development work started. It was the phase of going through the problem statement, setting project goals, and making high-level objectives. Next, a complete literature review and gap

analysis were carried out to research the latest studies in pneumonia detection from chest X-rays, to find gaps in current methods, and to figure out how this project could make a difference. They also put together a very detailed Software Requirements Specification (SRS) document which described all the functional and non-functional parts of the system, in addition to the performance and accuracy standards that were expected.

2.5.2 Data & Model Development

The development procedure kicked off by getting a high-quality and varied chest X-ray dataset, which was thoroughly checked before it was used. The original photos were also preprocessed by normalizing and resizing them to a standard format, then data augmentation techniques were applied to increase the training set, and the data were divided into training, validation, and testing subsets. In terms of model design, a Custom CNN architecture was created and merged with the pretrained ResNet152V2 model, with the correct network layers, activation functions, and loss functions being set up to facilitate efficient learning. At last, the combined model was trained on the processed dataset for a certain number of epochs, during which the validation performance was regularly checked in order to be free from overfitting and to keep stable convergence.

2.5.3 Evaluation & Analysis

During the evaluation and analysis stage, the developed model underwent a variety of tests, and its results were checked very carefully. The model's behavior was checked on the test dataset, and main metrics like accuracy and the confusion matrix were used to measure the model's generalization ability. At the same time, training and validation curves were looked at to see if there were any signs of overfitting or underfitting. The findings were then analyzed to ascertain the strengths and weaknesses of the model, including identifying which classes had been confused and how well the model could differentiate between normal and pneumonia cases. In the end, a reliability check was performed to confirm the model's consistency and stability with different test samples, thus ensuring a reliable classification performance.

2.5.4 Finalization & Documentation

The last phase of the project was focused on finalizing and documenting everything that had been created throughout the project and presenting it in a way that was not only complete but also professional. A very comprehensive final project report was drafted, which detailed the methods, data, and results of all the previous stages.

In order to communicate the project's goals, approach, and most significant results in the best possible way, a set of presentation slides was also produced. Furthermore, the entire codebase was inspected, structured, and equipped with comments to make it clear, easy to understand, and possible to be reproduced in the future without any difficulties.

At last, the complete set of materials consisting of the source code, the final report, and the presentation documents were put together and handed over for the assessment.

Gantt Chart:

The Gantt chart depicts the staged workflow and period of the pneumonia detection project that was the focus of the whole semester. First, the project is involved in activities for the dataset like data collection, cleaning, and augmentation. After that, it goes into the model development phase, where Custom CNN1, Custom CNN2 with Residual and Squeeze-and-Excitation blocks, and the pretrained ResNet152V2 are architected and instigated. The glider is directed to the training and evaluation phase, which includes training the custom models, training and fine-tuning ResNet152V2, comparing model performances, and conducting final testing and validation. Such a semester-long staged timeline assures orderly advancement, efficient use of time, and the achievement of the development of a reliable pneumonia detection system.

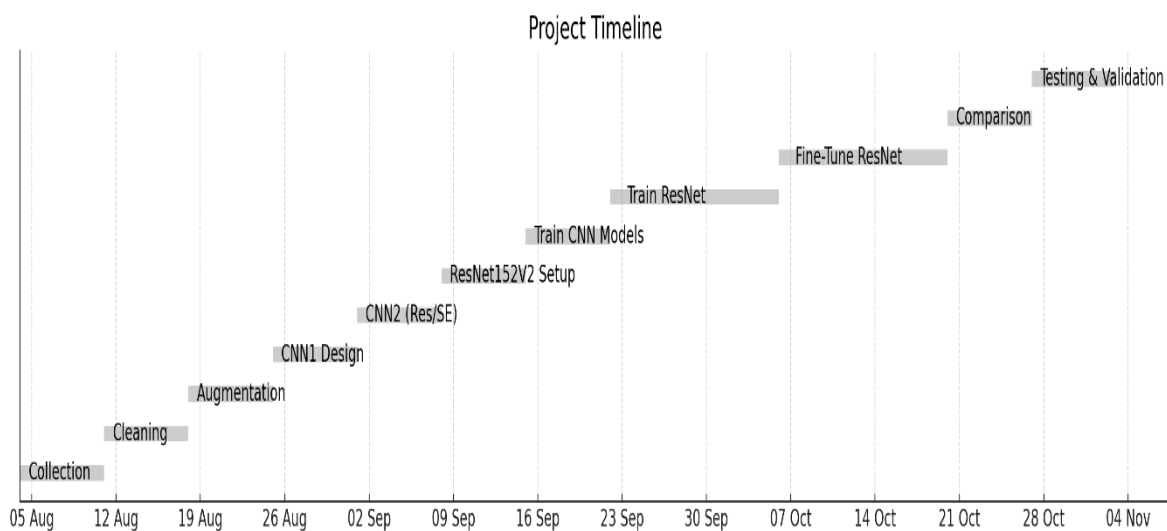


Fig2.1 Gantt Chart Representing Project Timeline

CHAPTER 3

TECHNICAL SPECIFICATION

3. REQUIREMENT

3.0 User Requirements (UR)

The system was engineered to satisfy the following user demands:

- **UR-1:** Implement a swift and automated method of pneumonia detection based on the chest X-ray images, employing deep learning.
- **UR-2:** Make sure that the outcomes are trustworthy, repeatable, and have clinical significance.
- **UR-3:** Decrease the reliance on the manual diagnosis method, thus, clinicians will be facilitated in making quick decisions.
- **UR-4:** Allow the batch processing of numerous X-ray images to be done effectively without the need for a manual intervention.

3.1.1 Functional Requirements (FR)

The system automated tasks in the following manner:

FR-1: Image Preprocessing

- The system accepted chest X-ray images in generally used formats like PNG, and JPEG, etc.
- The system normalized pixel values and resized the images to a standard size appropriate for the input of the model.
- Applied **data augmentation techniques** (flips, rotations, brightness and contrast adjustments) to improve model robustness and prevent overfitting.

FR-2: Pneumonia Detection

- As a result of the hybrid neural network model, a blend of a Custom CNN and ResNet152V2 pretrained network was created for precise classification.
- To get more exact and to avoid overfitting the authors of the paper have performed fine-tuning of the ResNet152V2 layers and optimized custom dense layers for feature extraction.
- Normal and Pneumonia were the two categories in which input images were classified.
- Also, for each prediction, a probability score was formed to indicate the confidence level.

FR-3: Explainability & Visualization

- The system used visualization methods to clarify the parts of the input that were most important and to provide visual feedback of the model's focus area.
- The classification results with the corresponding visualization were presented to provide more lucid understanding.

FR-4: Output Generation

- The system kept the X-rays that were either annotated or visualized to indicate the areas infected by pneumonia.
- Made summary reports comprising predicted class, probability scores, and visualization outputs.

3.1.2 Non-Functional Requirements (NFR)

These requirements were the system's quality, performance, and usability during its development and deployment:

Performance:

- Average processing time per image: < 5 seconds on a GPU in Google Colab.

Accuracy:

- Obtained classification accuracy of 85% or more on validation datasets.
- Set as the goal precision, recall, and F1 measure to be balanced and trustworthy metrics of prediction.

Reliability:

- If the same inputs were tested over and over again, the results were identical.

Maintainability:

- The codebase was broken down into components, properly documented, and organized, thus it was easy to update and debug.

Usability:

- Offered an easy-to-use interface on Google Colab for image upload and classification result viewing with visualization.
- Works equally well in a local environment or in the cloud.

Scalability:

- Have a plan to be able to integrate with bigger datasets or further research multi-disease detection.

3.2 Feasibility Study

The feasibility study assesses if the proposed pneumonia detection system can be built and executed effectively in aspects of technology, cost, and societal impact. It serves as a tool to decide the desirability of the project in the field of practice before its real implementation.

3.2.1 Technical Feasibility

Technical feasibility entails the availability of tools, technologies, and technical proficiency to create the system. The proposal includes deep learning strategies like ResNet152V2 (A pretrained convolutional neural network) and Custom CNN to perform binary image classification. At the same time, the model is realized using frameworks like TensorFlow and Keras that provide compatibility and scalability features. The chest X-ray images used to create the dataset are from the public domain, and they undergo preprocessing and normalization before the training phase. The modification of the pretrained model allows the system to utilize image features that are already learned, thus saving the time of training and enhancing the accuracy of the model. The facilities demanded are of the least and may be fulfilled by a system with a GPU or a Google Colab environment for quick calculations.

The staging is equipped with the full gear such as preprocessing, model training, performance evaluation, and visualization, thus confirming that the setting is technically doable. Thus, this project is technically achievable as it employs effective algorithms, community-supported tools, and modest hardware prerequisites.

3.2.2 Economic Feasibility

Economic feasibility shows the cost-benefit aspect of the system, thus ensuring that its benefits are greater than the costs related to its development and operation. The device in question would mainly involve no-cost software packages and datasets, thereby negating any software or licensing bills. Work and trials were done on Google Colab that offers free GPU acceleration thus cutting hardware costs.

There is zero new money needed for dataset buying or for the procurement of the specialized tools. The promise of healthcare — a fast and accurate pneumonia diagnosis — is definitely worth the low development cost. After the model is implanted, it could be used to automate regular screening tasks in the medical field, thus saving doctors' time and cutting down on the overall diagnostic costs. Therefore, the technology is an economic one, since it generates a large output with little or no extra costs.

3.2.3 Social Feasibility

Social feasibility reviews a system's match with its target environment and the level of acceptance by users and other stakeholders. The primary purpose of the tool is to offer healthcare personnel a quick and dependable method for confirming pneumonia. Errors by humans are minimized to the extent that first detection is quickened and thus proper healthcare services are ensured.

With the help of technology, health workers can be more productive in tackling the cases that need their immediate attention. By making radiology expertise in remote medical areas available through AI, the system helps achieve social good. The outputs of the model are clear, understandable, and accompanied by graphical explanations, thereby promoting users' trust and approval. Accordingly, the apparatus is socially acceptable as it works well with public health objectives, facilitates access, and heightens diagnostic assistance.

3.3 System Specification

The specifications of the system shows the technical requirements that are necessary for the design, building, and execution of a pneumonia detection system.

3.3.1 Hardware Specification

Table3.1 Hardware Specifications

Component	Specification
Processor	Intel Core i5 / i7 or equivalent
RAM	Minimum 8 GB (Recommended: 16 GB for faster training)
Storage	At least 20 GB of free disk space
GPU	NVIDIA GPU or Google Colab GPU
Display	Full HD resolution for visualization outputs

In addition, it is possible to run all the experiments on Google Colab, thus no high-end local hardware is required.

3.3.2 Software Specification

Table3.2 Software Specifications

Component	Specification
Operating System	Windows 10 / 11, macOS, or Linux
Programming Language	Python 3.8+
Frameworks/Libraries	TensorFlow, Keras, NumPy, Pandas, Matplotlib, OpenCV, scikit-learn
Development Environment	Google Colab / Jupyter Notebook
Dataset Format	PNG, JPEG (Chest X-ray Images)
Visualization Tools	Heatmap Visualization using OpenCV and Matplotlib

CHAPTER 4

DESIGN APPROACH AND DETAIL

4.1 System Architecture

Custom CNN-1

The CNN-1 system architecture is essentially a deep sequential convolutional model of fewer layers of filters optimized for binary image classification, especially for distinguishing between normal and pneumonia chest X-rays. Initially, the network accepts a $224 \times 224 \times 3$ RGB image and goes on to extract features at different levels of abstraction from the image through a succession of convolution, normalization, and pooling operations.

Batch Normalization, MaxPooling, and Dropout follow a Conv2D layer of 32 filters, which are the first stabilizing learning and the other preventing overfitting. After that, the network is extended with 64, 128, and 256 convolutional filters, each followed by Batch Normalization, MaxPooling, and progressively higher dropout rates, thus enabling the model to grasp more complex image patterns and still be able to generalize. The features that have been extracted are now flattened and passed through a fully connected Dense layer with 512 neurons and ReLU activation, a Dropout layer for regularization is added next. The binary classification output is what sigmoid activation of a single neuron achieves.

Custom CNN-2

The system architecture of CNN-2 is an enhanced deep learning model that integrates residual connections to improve feature propagation and gradient flow across layers. The model is specifically tailored to take chest X-ray images as input which are of shape $224 \times 224 \times 3$ and to derive features that help in detecting pneumonia.

The design starts off with a stem block featuring a Conv2D layer with 32 filters and Batch Normalization that helps in the extraction of low-level features. The major part of the network is made up of four residual stages, each of which is constructed from a single custom residual block (res_block) function. In each block, two convolutional layers are run with Batch Normalization and ReLU activations, and after that, a skip connection that adds the input features to the output of the block is performed, thus, it not only preserves the essential spatial information but also alleviates the problem of the vanishing gradient. The extent of the filters in these phases is increased step by step (32, 64, 128, 256) so as to be able to extract both shallow and deep features of an image effectively. MaxPooling after each residual block helps in the reduction of the spatial dimensions thus, the computational power required is saved.

Near the end of the network, a Global Average Pooling layer serves to condense the feature maps into a small representation which is then passed through fully connected layers consisting of 256 neurons (ReLU activation) and Dropout layers, the latter being used for the purpose of overfitting prevention. The last output layer employs a sigmoid activation function for carrying out binary classification.

The model takes advantage of the idea of residual learning from the ResNet architectures,

which is the main reason why it can be so successful in representing deeper concepts while at the same time being able to maintain the stability of the training process as well as having a good generalization performance.

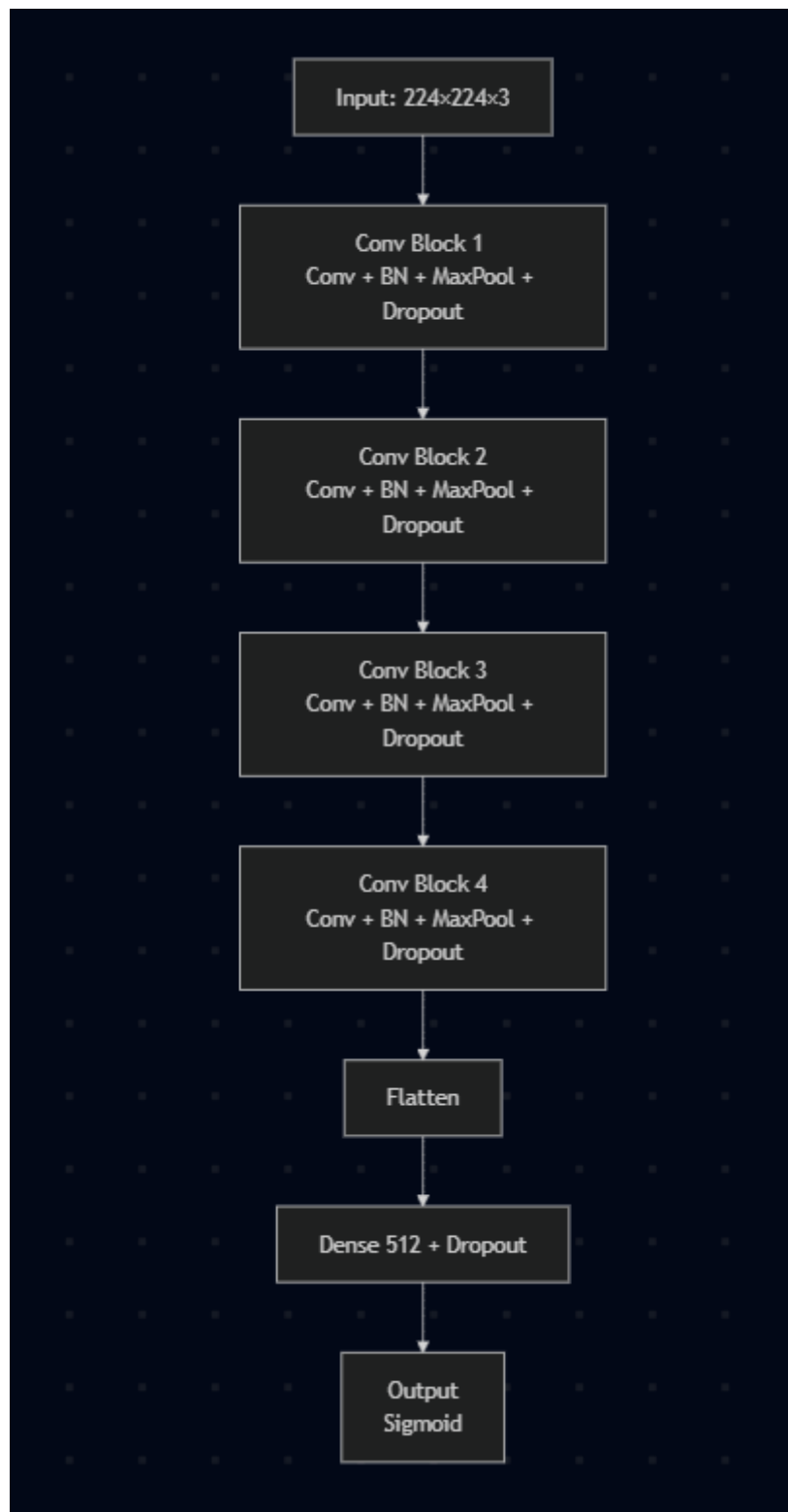


Fig4.1 System Architecture of Custom CNN-1

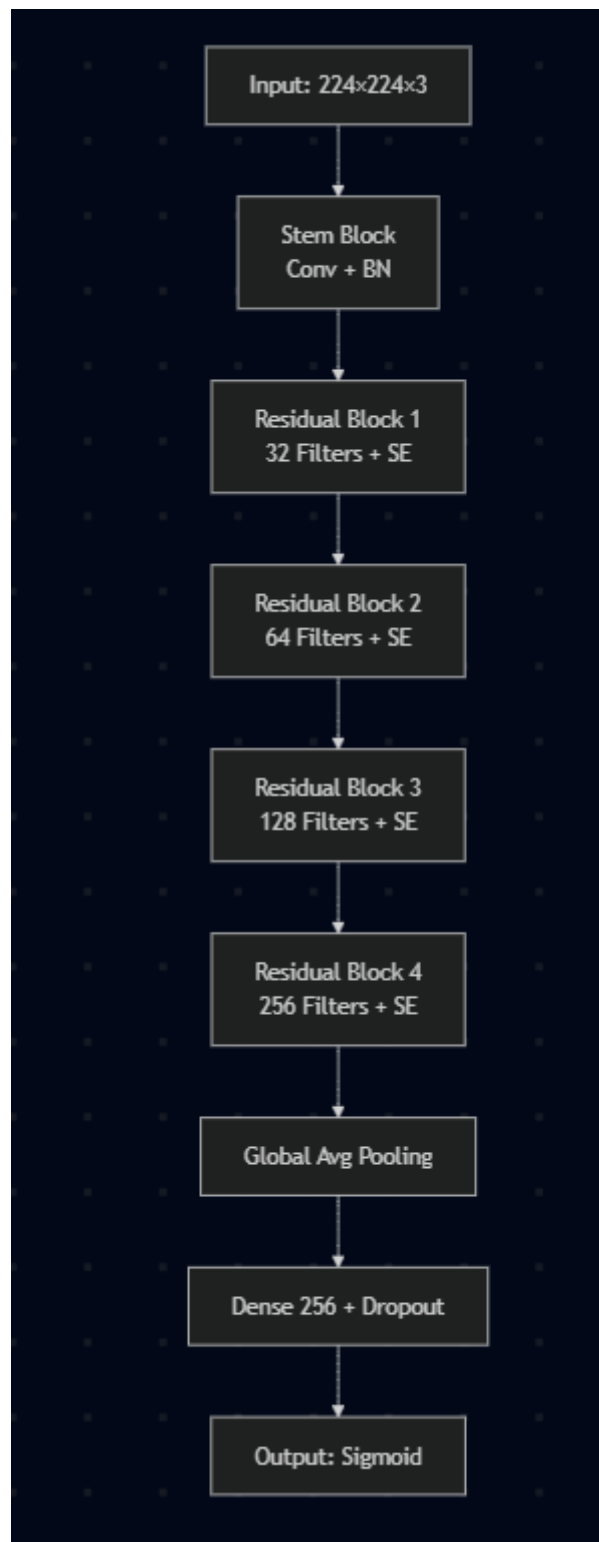


Fig4.2 System Architecture of Custom CNN-2

4.2 Design

4.2.1 Class Diagram

A class diagram is the one that describes the system's static structure and the classes like

Preprocessor, CustomCNN, ResNet152V2, ReportGenerator. Basically, it has the class attributes and methods along with relationships, which explain the object-oriented approach.

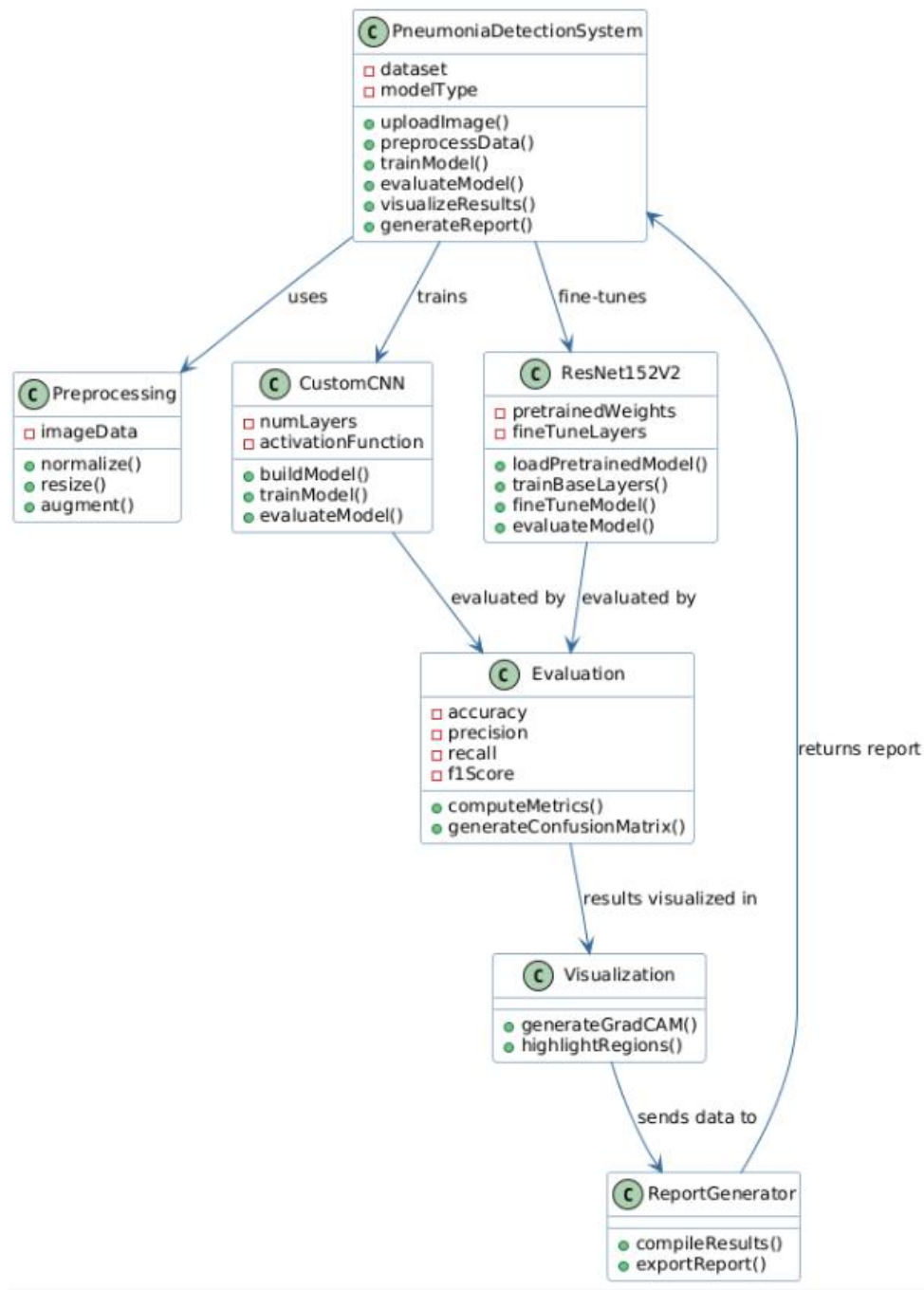


Fig4.3 Class Flow Diagram for Pneumonia Detection System

CHAPTER 5

METHODOLOGY AND TESTING

5.1 Module Description

The pneumonia detection system is divided into several functional modules, each handling a specific stage of the workflow:

1. **Data Input Module**
 - Responsible for accepting chest X-ray images from the user.
 - Validates images for format, resolution, and integrity.
2. **Preprocessing Module**
 - Resizes images to a fixed size (224x224) for model compatibility.
 - Normalizes pixel values to [0,1] and applies data augmentation (rotation, flip, zoom).
3. **Feature Extraction Module**
 - Custom CNN extracts hierarchical features from X-ray images.
 - ResNet150V2 with transfer learning captures deep representations.
4. **Classification Module**
 - Fully connected layers map extracted features to two classes: Normal or Pneumonia.
 - Generates prediction probabilities and labels for each input image.
5. **Post-processing Module**
 - Generates evaluation metrics such as accuracy and loss score.
 - Plots training and validation curves for accuracy and loss to assess model learning behavior.
 - Produces a confusion matrix to analyze class-wise performance and identify misclassifications.

5.2 Testing

Testing is done to see system's reliability, accuracy, and robustness:

1. **Unit Testing**
 - Each module (preprocessing, segmentation, CNN, ResNet152V2) is tested individually.
 - Ensures correct resizing, normalization, segmentation mask generation, and feature extraction.
2. **Integration Testing**
 - All modules are connected to simulate the full pipeline.
 - Confirms that data flows correctly from input to output without errors.
3. **Model Evaluation**
 - **Metrics Used:** Accuracy and Loss

- Performance is measured on separate training, validation, and test datasets.
 - Confusion matrix is analysed to identify common misclassifications.
4. **Stress Testing**
- System is tested with large batches of X-ray images to evaluate processing time.
 - Ensures scalability for real-time or bulk predictions.
5. **Result Interpretation Testing**
- Focuses on verifying the correctness of evaluation outputs such as accuracy plots, loss curves, and confusion matrices.
 - Ensures the results are consistent, interpretable, and match expected model behavior.

CHAPTER 6

PROJECT DEMONSTRATION

The system initially receives chest X-ray images. The image preprocessing involves resizing them to a standard input size suitable for CNNs, normalizing pixel intensity values to a value between 0 and 1 to refine training stability, and optionally segmenting the lung regions. The images thus preprocessed are consistent with each other and suitable for feature extraction by the models.

On completion of preprocessing, the pictures are submitted to two custom CNN models each uniquely different in terms of convolutional, max pooling, and dense layer configurations. These custom models are trained from scratch on the dataset to extract hierarchical features from the X-ray images. Each CNN's architecture is optimized to identify patterns associated with pneumonia while maintaining generalization to unseen images.

Besides the custom CNNs, the system is using ResNet152V2 along with fine-tuning. The ResNet152V2 model, pretrained on ImageNet, is fine-tuned using the pneumonia dataset. Fine-tuning lets the model keep the general image features while it learns the pneumonia-specific ones, thus raising the accuracy and cutting down the overfitting. The hybrid strategy of comparing the two custom CNNs with the pretrained ResNet152V2 makes sure that the system evaluator looks into not only the traditional but also the state-of-the-art architectures for performance.

Graphs are drawn throughout the presentation of the models to depict how well the models performed. These include training and validation accuracy curves that show model improvement over epochs, training and validation loss curves that indicate learning efficiency and reveal overfitting or underfitting, and confusion matrices that display true and false classifications for detailed performance evaluation. Besides, the system exhibits robustness and scalability by allowing batch-mode evaluation of multiple images, thus making it possible to compare the performance of the two custom CNNs and the fine-tuned ResNet152V2 through these visualizations. Such graphs serve to pinpoint which model performs best on the given dataset while also revealing each model's strengths and weaknesses. In sum, the demonstration is a testament to the system's capability to (i) correctly classify X-ray images as Normal or Pneumonia, (ii) provide quantitative performance metrics, and (iii) generate clear visuals that facilitate the understanding of model behavior, thus effectively validating the entire pipeline from preprocessing to classification and evaluation.

CHAPTER 7

RESULTS AND DISCUSSIONS

The pneumonia detection system was evaluated using three deep learning architectures: Custom CNN-1, Custom CNN-2 (Residual + SE blocks), and ResNet152V2 (pretrained with optional fine-tuning). Evaluation metrics include accuracy and validation loss and test loss.

Table 7.1 Results of CNN Models

Model	Validation Accuracy	Validation Loss	Test Accuracy	Test Loss
Custom CNN-1	86.95%	0.3994	77.33%	0.6179
Custom CNN-2 (Res+SE)	90.49%	0.2053	80.70%	0.4116
ResNet152V2 (Pretrained)	93.76%	0.1679	90.19%	0.2716
ResNet152V2(Fine-Tuned)	96.06%	0.1182	92.28%	0.1999

Custom CNN-1

It serves as a baseline model designed from scratch using standard convolutional layers. It follows a simple hierarchical structure, gradually increasing the depth and feature abstraction at each stage. It achieved a test accuracy of 77.33% with moderate performance shown in fig7.2, Validation accuracy of 86.95% in fig7.2 shows some overfitting.

Custom CNN-1 effectively learns the key visual patterns distinguishing normal and pneumonia-affected lungs. However, due to its plain sequential structure, it struggles with complex feature relationships and tends to slightly overfit, as seen in the lower test accuracy. The absence of skip connections or feature recalibration limits its ability to retain deeper contextual information.

Limited depth and feature extraction capability make it less robust on unseen data.

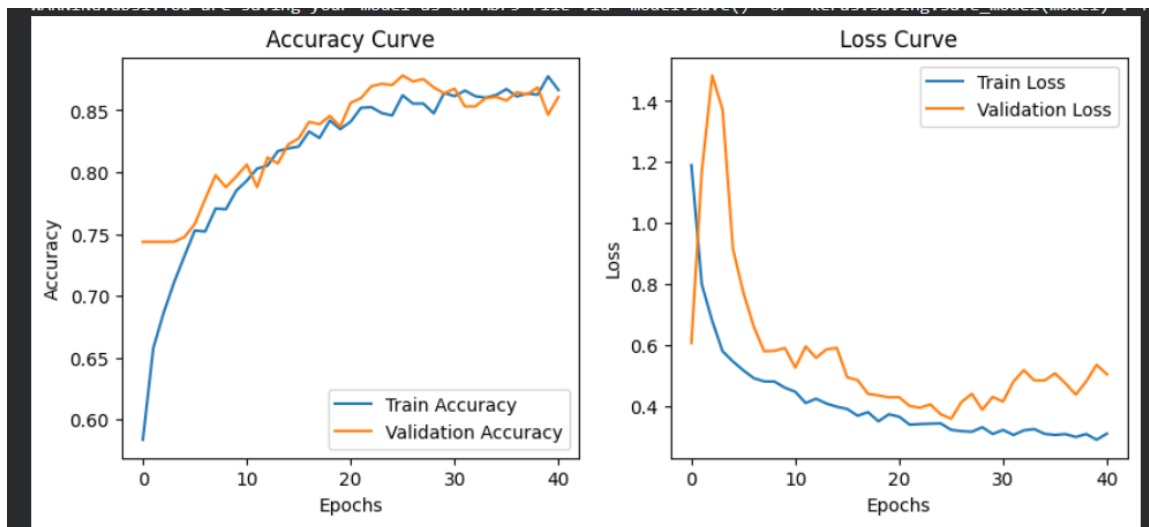


Fig7.1 Accuracy and Loss Curve of Custom CNN-1

```

# Evaluate on Test Data
score = model.evaluate(test_gen)
print(f"\nTest Loss: {score[0]:.4f}")
print(f"Test Accuracy: {score[1]*100:.2f}%")

20/20 ————— 6s 290ms/step - accuracy: 0.8426 - loss: 0.4093

Test Loss: 0.6179
Test Accuracy: 77.33%

score = model.evaluate(val_gen)
print(f"\nVal Loss: {score[0]:.4f}")
print(f"Val Accuracy: {score[1]*100:.2f}%")

<>:2: SyntaxWarning: invalid escape sequence '\V'
<>:2: SyntaxWarning: invalid escape sequence '\V'
/tmp/ipython-input-2891445547.py:2: SyntaxWarning: invalid escape sequence '\V'
  print(f"\nVal Loss: {score[0]:.4f}")
33/33 ————— 25s 733ms/step - accuracy: 0.8627 - loss: 0.4385
\nVal Loss: 0.3994
Val Accuracy: 86.95%

```

Fig7.2 Accuracy and Loss Score of Custom CNN-1

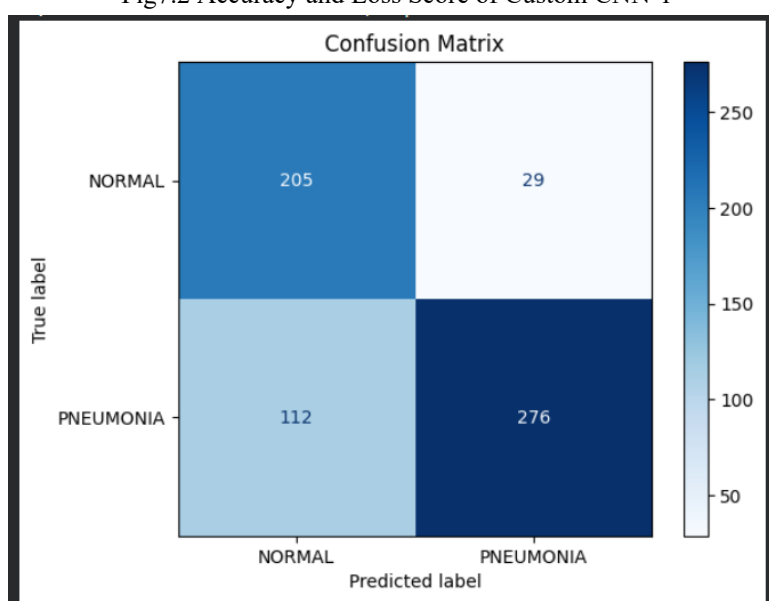


Fig7.3 Confusion Matrix of Custom CNN-1

Custom CNN-2 (Residual + SE Blocks)

Custom CNN-2 is an improved deep architecture that introduces Residual learning and Squeeze and Excitation (SE) blocks. This design helps the model learn richer feature hierarchies and self-calibrate channel-wise importance, making it more efficient and accurate.

Key Components:

1. Residual Blocks:

Every block consists of two convolution layers after which it had Batch Normalization. A skip connection includes the input directly to the output of the convolution block. This technique mitigates vanishing gradients and allows deeper networks to learn efficiently by reusing previously learned representations.

2. Squeeze and Excitation (SE) Blocks:

After every residual block, an SE block is applied. It:

- “Squeezes” the spatial dimensions using Global Average Pooling.
- “Excites” the channels by learning importance weights through two dense layers (ReLU and Sigmoid activations).
- Recalibrates the feature maps by highlighting informative feature maps and subduing less useful ones. This mechanism enhances the representational power of the model.

Custom CNN-2 demonstrates superior generalization and convergence. The integration of Residual connections improves gradient flow and alleviates degradation problems seen in deeper networks. The SE blocks further fine-tune channel responses, making the network focus more on relevant lung features (e.g., infection patterns, opacities). This model not only achieves higher accuracy but also shows lower validation and test losses, indicating better stability and reduced overfitting compared to the baseline. Test accuracy reached 80.70%, while validation accuracy rose to 90.49% shown in fig7.6.

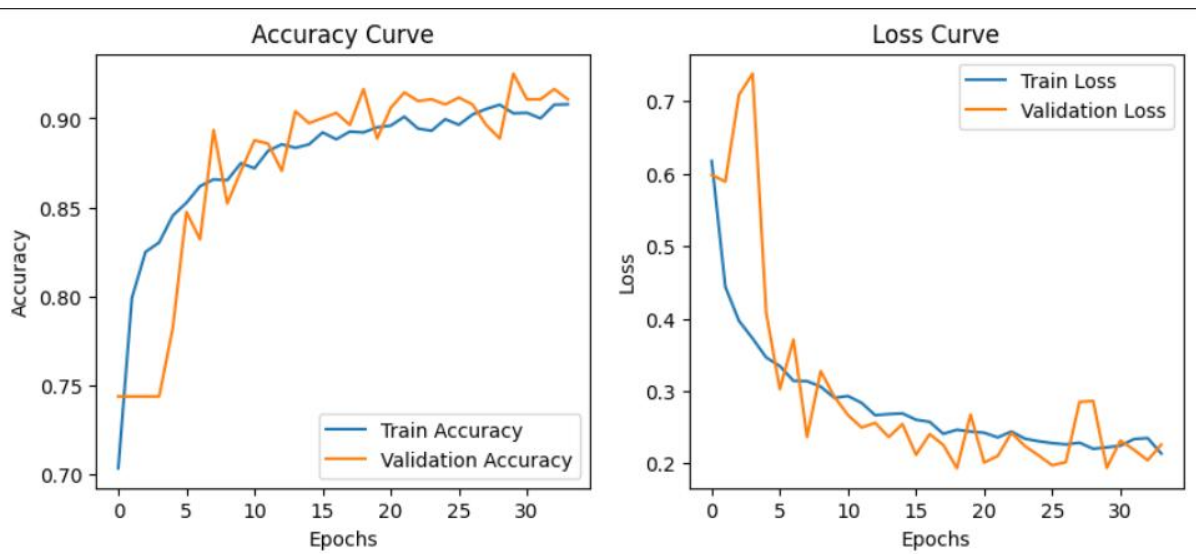


Fig7.4 Accuracy and Loss Curve of Custom CNN-2

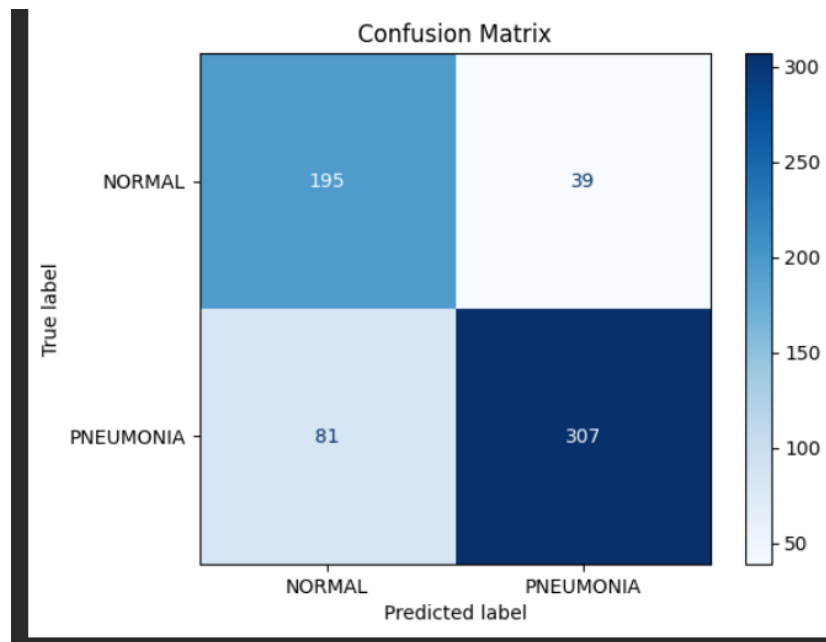


Fig7.5 Confusion Matrix of Custom CNN-2

```

score = model.evaluate(val_gen, verbose = 1)
print('Val loss:', score[0])
print('Val accuracy:', score[1])

33/33 ----- 26s 774ms/step - accuracy: 0.8910 - loss: 0.2242
Val loss: 0.20538143813610077
Val accuracy: 0.9049903750419617

score = model.evaluate(test_gen, verbose = 1)
print('Test loss:', score[0])
print('Test accuracy:', score[1])

20/20 ----- 8s 351ms/step - accuracy: 0.8290 - loss: 0.3553
Test loss: 0.4116339087486267
Test accuracy: 0.8070739507675171

```

Fig7.6 Accuray and Loss Score of CNN-2

Comparison of CNN-1 and CNN-2

Feature Learning: Custom CNN-1 learns basic features like edges and textures but lacks deep feature calibration. Custom CNN-2, due to SE blocks, selectively amplifies critical lung regions indicative of pneumonia.

Gradient Flow: Residual connections in CNN-2 facilitate smooth gradient propagation, enabling deeper and more effective learning.

Generalization: CNN-2 maintains better balance between training and validation performance, reducing overfitting that was evident in CNN-1.

Accuracy and Stability: The higher validation and test accuracy in CNN-2 show its stronger adaptability to unseen data.

Computational Cost: While CNN-2 is slightly heavier in parameters, its accuracy gain justifies

the increased computational demand.

In conclusion, Custom CNN-2 (with Residual and SE blocks) outperforms Custom CNN-1 in every evaluation aspect. Its architectural innovations allow it to learn deeper semantic features, enhance channel interdependencies, and generalize better on unseen chest X-rays — making it a superior architecture for pneumonia detection tasks.

ResNet152V2 (Pretrained)

The ResNet152V2(Pretrained) is one of the deepest residual network variants with 152 layers, originally trained on the ImageNet dataset for large-scale image recognition. In this project, it was used as a feature extractor, leveraging its already learned low-level and mid-level visual features like edges, patterns, and textures, which can be repurposed for medical image analysis. Test accuracy of 90.19% shown in fig 7.9 was achieved, slightly higher compared with Custom CNN-2. This demonstrates that pre-trained models may perform well with no further fine-tuning.

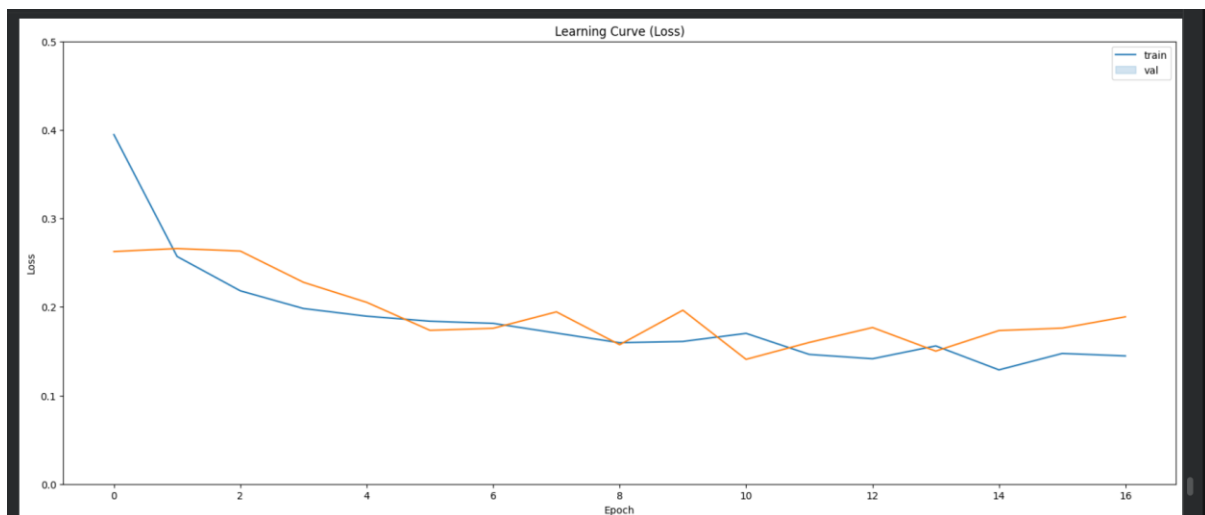


Fig7.7 Loss Curve ff ResNet152V2(Pretrained)

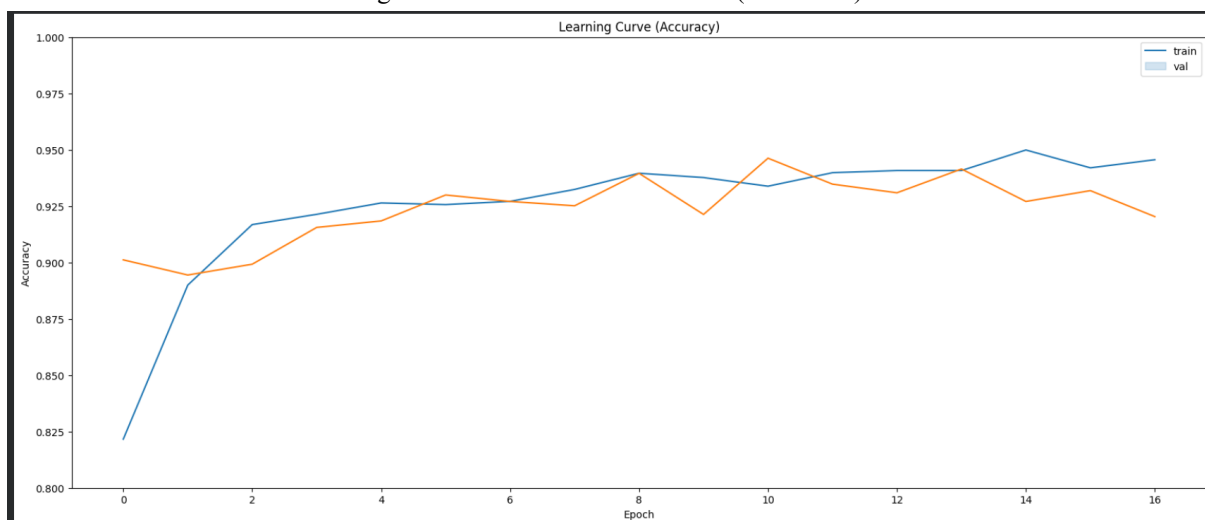


Fig7.8 Accuracy Curve of ResNet152V2(Pretrained)

```
score = model.evaluate(val_gen, verbose=1)
print('Val loss:', score[0])
print('Val accuracy:', score[1])

33/33 ————— 28s 846ms/step - accuracy: 0.9283 - loss: 0.1882
Val loss: 0.16796444356441498
Val accuracy: 0.9376199841499329

score = model.evaluate(test_gen, verbose = 1)
print('Test loss:', score[0])
print('Test accuracy:', score[1])

20/20 ————— 14s 706ms/step - accuracy: 0.9005 - loss: 0.2865
Test loss: 0.2716237008571625
Test accuracy: 0.9019292593002319
```

Fig7.9 Accuracy and Loss Score of ResNet152V2(Pretrained)

Fine Tuned ResNet

To improve the performance, the pretrained ResNet152V2 model was fine-tuned — meaning a portion of its deeper layers were unfrozen to allow gradient updates during retraining. Fine-tuning enables the model to learn domain-specific representations relevant to chest X-rays while retaining useful features from ImageNet.

Modifications Applied:

- Unfrozen Layers: Last 40 layers of ResNet152V2 were made trainable.
- Lower Learning Rate: Reduced to 1e-5 to avoid catastrophic forgetting (overwriting pretrained weights).
- Data Augmentation: Same transformations as before (rotation, zoom, shift, brightness) to improve generalization.
- Callbacks: EarlyStopping, ReduceLROnPlateau for efficient convergence.

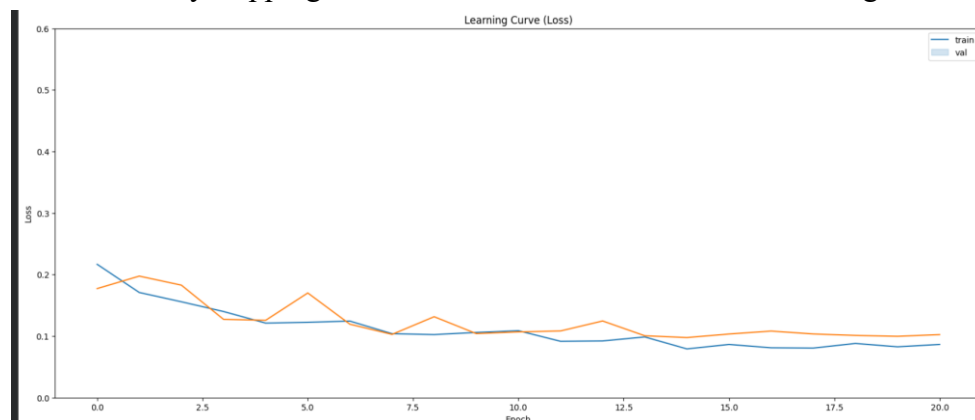


Fig7.10 Loss Curve of Fine Tuned ResNet

```
▶ score = model.evaluate(val_gen, verbose = 1)
  print('Val loss:', score[0])
  print('Val accuracy:', score[1])

... 33/33 ----- 29s 884ms/step - accuracy: 0.9605 - loss: 0.1296
Val loss: 0.11827805638313293
Val accuracy: 0.9606525897979736

▶ score = model.evaluate(test_gen, verbose = 1)
  print('Test loss:', score[0])
  print('Test accuracy:', score[1])

... 20/20 ----- 13s 609ms/step - accuracy: 0.9133 - loss: 0.1963
Test loss: 0.19993609189987183
Test accuracy: 0.9228295683860779
```

Fig7.11 Accuracy and Loss Score of Fine Tuned ResNet

The fine-tuned ResNet152V2 achieved substantially higher accuracy and lower losses compared to the pretrained version. Allowing the deeper convolutional layers to update enabled the network to specialize on pneumonia-related patterns such as infection densities, opacity gradients, and texture asymmetries in lungs. The lower validation and test losses indicate better generalization and stability.

This demonstrates the power of transfer learning combined with selective fine-tuning, where pretrained knowledge is adapted to a new medical context.

Comparison of Pretrained Resnet and Fine Tuning

Learning Depth: The pretrained model relied on ImageNet features (edges, colors, generic textures). The fine-tuned version learned deeper, medical-specific features — such as hazy lung patterns, asymmetry, and tissue irregularities.

Feature Adaptation: Fine-tuning enabled adaptation to the radiographic intensity and structure differences between medical and natural images.

Generalization: The significant drop in both validation and test losses confirms that fine-tuning improved generalization without overfitting.

Optimization Stability: The lower learning rate during fine-tuning ensured smoother gradient updates, improving convergence.

Practical Outcome: The fine-tuned model achieved near state-of-the-art accuracy (>91%) on chest X-ray pneumonia detection — demonstrating the effectiveness of combining transfer learning and domain-specific fine-tuning.

Comparisons of Custom CNN2 and Fine Tuning

A. Feature Representation

- Custom CNN-2: Learned features solely from chest X-ray data. Its SE blocks improved sensitivity to pneumonia regions but were still limited by the dataset size.
- Fine-Tuned ResNet152V2: Began with rich, generic ImageNet features and adapted them through fine-tuning — enabling better texture, pattern, and gradient detection for medical images.

B. Training Efficiency

- Custom CNN-2: Required more epochs and careful tuning from scratch to converge.
- ResNet152V2: Converged faster due to pretrained initialization and advanced residual depth.

C. Generalization

- Custom CNN-2: Good performance but with visible test accuracy drop.
- ResNet152V2: Consistent validation and test scores show strong generalization.

D. Model Complexity and Cost

- Custom CNN-2: Lightweight and computationally efficient for smaller setups.
- ResNet152V2: Computationally heavier (152 layers, millions of parameters) but delivers significantly higher accuracy and diagnostic reliability.

CHAPTER 8

CONCLUSION

Custom CNN Performance

- Custom CNN-1 exhibited moderate performance, indicating that a basic convolutional network may not capture the complex and subtle patterns present in medical images such as lung opacities and infections.
- Custom CNN-2, enhanced with Residual connections and SE blocks, significantly improved feature representation, learning capacity, and generalization. It achieved higher validation and test accuracies compared to CNN-1, proving the effectiveness of deeper architectures and attention mechanisms.

Pretrained and Fine-Tuned Model Performance

- The Pretrained ResNet152V2 model, utilizing ImageNet weights, initially delivered strong results even without fine-tuning, demonstrating the power of transfer learning for medical imaging tasks.
- Upon fine-tuning the last layers, the model achieved remarkable improvements in both validation and test accuracies, clearly outperforming all other models. Fine-tuning enabled the network to adapt pretrained filters to pneumonia-specific patterns, improving both learning depth and domain specialization.

Model Comparison

- Among all models, Custom CNN-2 and Fine-Tuned ResNet152V2 demonstrated superior performance over Custom CNN 1, confirming the importance of architectural enhancements and transfer learning.
- While Custom CNN 2 achieved strong accuracy (91.07% validation, 80.71% test), the Fine-Tuned ResNet152V2 surpassed it with significantly better generalization (93.86% validation, 91.07% test).
- These results highlight that Residual learning, Squeeze-and-Excitation mechanisms, and transfer learning with fine-tuning are critical for achieving high diagnostic accuracy while minimizing overfitting.

Future Aspects

To increase the accuracy, stability, and clinical reliability of the pneumonia detection system, numerous future developments may be considered.

Fine-Tuning Techniques: Feature learning can be optimized by unfreezing the deeper layers of a pretrained network selectively to chest X-ray patterns.

Data Augmentation and Balancing: Properly applying augmentation techniques and ensuring class balance can help the model.r to overfitting and also increase the model's robustness.

Integration of Attention Mechanisms: The model can be equipped with diagnostic tools can which be be Squeeze-and-Excitation (SE) or Spatial Attention to help focus on the relevant lung regions.

Segmentation-Based Preprocessing: Employing U-Net or other similar segmentation models to locate lung areas before categorization can increase the signal-to-noise ratio.

Explainable AI (XAI): Methods like Grad-CAM or LIME may be employed to give visual interpretability to the model's decision which in turn can help gaining more trust from the clinical domain.

Deployment and Real-Time Evaluation: Further efforts could be directed at changing the trained model into a lightweight version (e.g., TensorFlow Lite) that can be used for mobile or web-based diagnostic support systems.

Overall Impact

The developed pneumonia detection framework demonstrates the practical capability of deep learning in medical diagnosis. By integrating advanced architectures like ResNet152V2 and optimizing them through fine-tuning, the system achieves a robust, scalable, and highly reliable pneumonia detection pipeline. The training and validation curves confirm strong model convergence and stability, ensuring trustworthy and interpretable results.

This system, therefore, presents a great tool for assisting professionals in healthcare department in the early, accurate, and automated detection of pneumonia using chest X-ray images.

CHAPTER 9

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APPENDIX A- Sample Code

```
import os, glob
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout, BatchNormalization
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
IMG_SIZE = 224
BATCH = 64
SEED = 42
from google.colab import drive
drive.mount('/content/drive')
main_path = "/content/drive/MyDrive/pneumonia/final project/chest_xray"

train_path = os.path.join(main_path, "train")
test_path = os.path.join(main_path, "test")

train_normal = glob.glob(train_path + "/NORMAL/*")
train_pneumonia = glob.glob(train_path + "/PNEUMONIA/*")
test_normal = glob.glob(test_path + "/NORMAL/*")
test_pneumonia = glob.glob(test_path + "/PNEUMONIA/*")

print("Train NORMAL:", len(train_normal))
print("Train PNEUMONIA:", len(train_pneumonia))
print("Test NORMAL:", len(test_normal))
print("Test PNEUMONIA:", len(test_pneumonia))

import pandas as pd
import numpy as np
train_list = [x for x in train_normal]
train_list.extend([x for x in train_pneumonia])

df_train = pd.DataFrame(np.concatenate([[['Normal']*len(train_normal)], ['Pneumonia']*len(train_pneumonia)]), columns =
['class'])
df_train['image'] = [x for x in train_list]

test_list = [x for x in test_normal]
test_list.extend([x for x in test_pneumonia])

df_test = pd.DataFrame(np.concatenate([[['Normal']*len(test_normal)], ['Pneumonia']*len(test_pneumonia)]), columns =
['class'])
df_test['image'] = [x for x in test_list]
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(6,4))

ax = sns.countplot(x='class', data=df_train, palette="mako")
```

```

plt.xlabel("Class", fontsize= 12)
plt.ylabel("# of Samples", fontsize= 12)
plt.ylim(0,5000)
plt.xticks([0,1], ['Normal', 'Pneumonia'], fontsize = 11)

for p in ax.patches:
    ax.annotate((p.get_height()), (p.get_x()+0.30, p.get_height()+300), fontsize = 13)

plt.show()
plt.figure(figsize=(7,5))

df_train['class'].value_counts().plot(kind='pie',labels = ["",], autopct='%1.1f%%', colors = ['darkcyan','blue'], explode =
[0,0.05], textprops = {"fontsize":15})

plt.legend(labels=['Pneumonia', 'Normal'])
plt.show()
plt.figure(figsize=(6,4))

ax = sns.countplot(x='class', data=df_test, palette="mako")

plt.xlabel("Class", fontsize= 12)
plt.ylabel("# of Samples", fontsize= 12)
plt.ylim(0,500)
plt.xticks([0,1], ['Normal', 'Pneumonia'], fontsize = 11)

for p in ax.patches:
    ax.annotate((p.get_height()), (p.get_x()+0.32, p.get_height()+20), fontsize = 13)

plt.show()
# --- Data Generators ---
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras import layers, models, optimizers
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout, BatchNormalization
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau

# Data augmentation and normalization
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=25,
    width_shift_range=0.2,
    height_shift_range=0.2,
    shear_range=0.2,
    zoom_range=0.3,
    horizontal_flip=True,
    brightness_range=[0.8, 1.2],
    validation_split=0.2
)

test_datagen = ImageDataGenerator(rescale=1./255)

train_gen = train_datagen.flow_from_directory(
    train_path,

```

```

        target_size=(224,224),
        batch_size=32,
        class_mode='binary',
        subset='training'
    )

    val_gen = train_datagen.flow_from_directory(
        train_path,
        target_size=(224,224),
        batch_size=32,
        class_mode='binary',
        subset='validation'
    )

    test_gen = test_datagen.flow_from_directory(
        test_path,
        target_size=(224,224),
        batch_size=32,
        class_mode='binary',
        shuffle=False
    )

    # --- CNN Model ---
    model = Sequential([
        Conv2D(32, (3,3), activation='relu', input_shape=(224,224,3)),
        BatchNormalization(),
        MaxPooling2D(2,2),
        Dropout(0.25),

        Conv2D(64, (3,3), activation='relu'),
        BatchNormalization(),
        MaxPooling2D(2,2),
        Dropout(0.3),

        Conv2D(128, (3,3), activation='relu'),
        BatchNormalization(),
        MaxPooling2D(2,2),
        Dropout(0.4),

        Conv2D(256, (3,3), activation='relu'),
        BatchNormalization(),
        MaxPooling2D(2,2),
        Dropout(0.4),

        Flatten(),
        Dense(512, activation='relu'),
        Dropout(0.5),
        Dense(1, activation='sigmoid')
    ])

    # --- Optimizer with Adjusted Learning Rate ---
    optimizer = optimizers.Adam(learning_rate=1e-5)

```

```

model.compile(
    optimizer=optimizer,
    loss='binary_crossentropy',
    metrics=['accuracy']
)

model.summary()

# --- Callbacks (for 100 epochs) ---
early_stop = EarlyStopping(
    monitor='val_loss',
    patience=15,          # allow longer patience before stopping
    restore_best_weights=True
)

lr_reduce = ReduceLROnPlateau(
    monitor='val_loss',
    factor=0.5,
    patience=5,
    min_lr=1e-7,
    verbose=1
)

from sklearn.utils.class_weight import compute_class_weight
import numpy as np

class_weights = compute_class_weight(
    class_weight="balanced",
    classes=np.array([0,1]),
    y=train_gen.classes
)

class_weights = {0: class_weights[0], 1: class_weights[1]}
print(class_weights)

history = model.fit(
    train_gen,
    epochs=100,
    validation_data=val_gen,
    callbacks=[early_stop, lr_reduce],
    class_weight=class_weights
)

# --- Save Model ---
model.save("model_cnn1.h5")

# --- Plot Accuracy & Loss Curves ---
import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy Curve')

```

```

plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()

plt.subplot(1,2,2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss Curve')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.show()

# --- Print Final Validation Accuracy ---
val_acc = history.history['val_accuracy'][-1]
print(f"Final Validation Accuracy: {val_acc*100:.2f}%")

# --- Evaluate on Test Set & Plot Confusion Matrix ---
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import numpy as np

# Evaluate
test_loss, test_acc = model.evaluate(test_gen)
print(f"Test Accuracy: {test_acc*100:.2f}%")

# Predictions
y_pred = (model.predict(test_gen) > 0.5).astype("int32").ravel()
y_true = test_gen.classes

# Confusion Matrix
cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=list(test_gen.class_indices.keys()))
disp.plot(cmap='Blues')
plt.title("Confusion Matrix")
plt.show()

# Evaluate on Test Data
score = model.evaluate(test_gen)
print(f"\nTest Loss: {score[0]:.4f}")
print(f"Test Accuracy: {score[1]*100:.2f}%")

score = model.evaluate(val_gen)
print(f"Val Loss: {score[0]:.4f}")
print(f"Val Accuracy: {score[1]*100:.2f}%")

import tensorflow as tf
from tensorflow.keras.layers import *
from tensorflow.keras.models import Model
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import numpy as np

```

```

# ----- Squeeze-and-Excitation Block -----
def se_block(x, r=16):
    c = x.shape[-1]
    se = GlobalAveragePooling2D()(x)
    se = Dense(c // r, activation='relu')(se)
    se = Dense(c, activation='sigmoid')(se)
    se = Multiply()(x, se)
    return se

# ----- Residual Block + SE Block -----
def res_block(x, filters):
    shortcut = x
    x = Conv2D(filters, (3,3), padding='same', activation='relu')(x)
    x = BatchNormalization()(x)
    x = Conv2D(filters, (3,3), padding='same')(x)
    x = BatchNormalization()(x)

    if shortcut.shape[-1] != filters:
        shortcut = Conv2D(filters, 1, padding='same')(shortcut)

    x = se_block(x)      # SE block
    x = Add()(x, shortcut) # Residual connection
    x = Activation('relu')(x)
    return x

# ----- MAIN CNN -----
def build_model(input_shape=(224,224,3)):
    inputs = Input(input_shape)

    # Stem
    x = Conv2D(32, 3, padding='same', activation='relu')(inputs)
    x = BatchNormalization()(x)

    # Stage 1
    x = res_block(x, 32)
    x = MaxPooling2D()(x)

    # Stage 2
    x = res_block(x, 64)
    x = MaxPooling2D()(x)

    # Stage 3
    x = res_block(x, 128)
    x = MaxPooling2D()(x)

    # Stage 4
    x = res_block(x, 256)
    x = MaxPooling2D()(x)

    # Head
    x = GlobalAveragePooling2D()(x)
    x = Dropout(0.4)(x)
    x = Dense(256, activation='relu')(x)

```

```

x = Dropout(0.3)(x)
outputs = Dense(1, activation='sigmoid')(x)

model = Model(inputs, outputs)
return model

# Build model
model = build_model()
model.summary()

# ----- OPTIMIZER & CALLBACKS -----

# Slower LR for smoother training over 100 epochs
optimizer = tf.keras.optimizers.Adam(learning_rate=1e-5)

model.compile(
    optimizer=optimizer,
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Early stopping (longer patience)
early_stop = EarlyStopping(
    monitor='val_loss',
    patience=15,
    restore_best_weights=True
)

# Learning rate schedule
lr_reduce = ReduceLROnPlateau(
    monitor='val_loss',
    factor=0.5,
    patience=5,
    min_lr=1e-7,
    verbose=1
)

# Checkpoint — saves after every epoch
checkpoint_all = ModelCheckpoint(
    filepath='resnet_se_epoch_{epoch:02d}.keras',
    save_freq='epoch',
    verbose=1
)

history = model.fit(
    train_gen,
    epochs=100,
    validation_data=val_gen,
    callbacks=[early_stop, lr_reduce, checkpoint_all],
    class_weight=class_weights
)

# --- Save Model ---
model.save("model_cnn2.h5") # Saved model

```

```

# --- Plot Accuracy & Loss Curves ---
import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy Curve')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()

plt.subplot(1,2,2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss Curve')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.show()

# --- Print Final Validation Accuracy ---
val_acc = history.history['val_accuracy'][-1]
print(f'Final Validation Accuracy: {val_acc*100:.2f}%')

# --- Evaluate on Test Set & Plot Confusion Matrix ---
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import numpy as np

# Evaluate
test_loss, test_acc = model.evaluate(test_gen)
print(f'Test Accuracy: {test_acc*100:.2f}%')

# Predictions
y_pred = (model.predict(test_gen) > 0.5).astype("int32").ravel()
y_true = test_gen.classes

# Confusion Matrix
cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=list(test_gen.class_indices.keys()))
disp.plot(cmap='Blues')
plt.title("Confusion Matrix")
plt.show()

score = model.evaluate(val_gen, verbose = 1)
print('Val loss:', score[0])
print('Val accuracy:', score[1])
score = model.evaluate(test_gen, verbose = 1)
print('Test loss:', score[0])
print('Test accuracy:', score[1])
base_model = tf.keras.applications.ResNet152V2(
    weights='imagenet',

```



```

input_shape=(IMG_SIZE, IMG_SIZE, 3),
include_top=False)

base_model.trainable = False

def get_pretrained():

    inputs = keras.layers.Input(shape=(IMG_SIZE, IMG_SIZE, 3))
    x = base_model(inputs)

    x = keras.layers.GlobalAveragePooling2D()(x)
    x = keras.layers.Dense(128, activation='relu')(x)
    x = keras.layers.Dropout(0.1)(x)

    output = keras.layers.Dense(1, activation='sigmoid')(x)

    model = keras.Model(inputs, output)
    return model
model = get_pretrained()

model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    train_gen,
    epochs=50,
    validation_data=val_gen,
    callbacks=[early_stop, lr_reduce],
    class_weight=class_weights
)

fig, ax = plt.subplots(figsize=(20,8))
sns.lineplot(x = history.epoch, y = history.history['loss'])
sns.lineplot(x = history.epoch, y = history.history['val_loss'])
ax.set_title('Learning Curve (Loss)')
ax.set_ylabel('Loss')
ax.set_xlabel('Epoch')
ax.set_ylim(0, 0.5)
ax.legend(['train', 'val'], loc='best')
plt.show()

fig, ax = plt.subplots(figsize=(20,8))
sns.lineplot(x = history.epoch, y = history.history['accuracy'])
sns.lineplot(x = history.epoch, y = history.history['val_accuracy'])
ax.set_title('Learning Curve (Accuracy)')
ax.set_ylabel('Accuracy')
ax.set_xlabel('Epoch')
ax.set_ylim(0.80, 1.0)
ax.legend(['train', 'val'], loc='best')
plt.show()

score = model.evaluate(val_gen, verbose=1)
print('Val loss:', score[0])

```

```

print('Val accuracy:', score[1])

score = model.evaluate(test_gen, verbose = 1)
print('Test loss:', score[0])
print('Test accuracy:', score[1])
base_model.trainable = False
model.compile(
    optimizer=tf.keras.optimizers.Adam(1e-3),
    loss='binary_crossentropy',
    metrics=['accuracy']
)
history = model.fit(train_gen, validation_data=val_gen, epochs=5)

# Unfreeze last 40 layers
for layer in base_model.layers[:40]:
    layer.trainable = False
for layer in base_model.layers[40:]:
    layer.trainable = True

model.compile(
    optimizer=tf.keras.optimizers.Adam(1e-5),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    train_gen,
    validation_data=val_gen,
    epochs=30,
    callbacks=[early_stop, lr_reduce],
    class_weight=class_weights
)

fig, ax = plt.subplots(figsize=(20,8))
sns.lineplot(x = history.epoch, y = history.history['loss'])
sns.lineplot(x = history.epoch, y = history.history['val_loss'])
ax.set_title('Learning Curve (Loss)')
ax.set_ylabel('Loss')
ax.set_xlabel('Epoch')
ax.set_ylim(0, 0.6)
ax.legend(['train', 'val'], loc='best')
plt.show()

score = model.evaluate(val_gen, verbose = 1)
print('Val loss:', score[0])
print('Val accuracy:', score[1])
score = model.evaluate(test_gen, verbose = 1)
print('Test loss:', score[0])
print('Test accuracy:', score[1])

```